

RESEARCH ARTICLE

KC-UNet: Enhancing U-Net With KAN and CBAM for Medical Image Segmentation

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ABSTRACT Medical image segmentation is a critical task in medical image analysis. However, traditional convolutional neural network (CNN) based methods are limited in modeling long-range dependencies, while Transformer-based segmentation models, though effective, suffer from high computational complexity due to quadratic attention operations. To address these challenges, this paper proposes an innovative U-Net variant, KC-UNet, which integrates Kolmogorov-Arnold Networks (KAN) and the Channel-Spatial Attention Module (CBAM). KC-UNet leverages the KAN representation theorem to represent features more efficiently, while CBAM enhances the model's ability to adaptively capture both spatial and channel-wise dependencies, striking a balance between accuracy and computational efficiency. To validate the effectiveness of CBAM, this paper conducts comprehensive ablation experiments by replacing CBAM with Squeeze-and-Excitation (SE), and Efficient Channel Attention (ECA), as well as removing the attention module entirely. Results demonstrate that CBAM provides the most significant performance improvements in terms of segmentation accuracy, confirming its superior capability in enhancing feature representation. This study evaluates KC-UNet on four widely used benchmark datasets (BUSI, GLAS, CVC, and ISIC2017) and compare it against recent state-of-the-art models such as TransUNet, Swin-unet, and U-KAN. KC-UNet consistently achieves the best performance, with an IoU of 66.60% on BUSI, outperforming Swin-unet by 1.28%, and a Dice score of 80.46%, which improves upon the baseline U-Net by 7.44%. Similar advantages are observed on GLAS and ISIC2017, demonstrating the effectiveness and generalizability of our approach across different modalities. To the best of our knowledge, KC-UNet is the first framework to integrate KAN and CBAM for medical image segmentation.

INDEX TERMS CBAM, KAN, medical image segmentation, U-Net.

I. INTRODUCTION

Medical image segmentation is an essential component of medical image analysis, aiming to delineate various anatomical structures or pathological regions with high precision [1], [2]. It holds significant importance in supporting clinical diagnosis, surgical planning, and therapeutic assessment [3], [4]. Earlier segmentation techniques primarily depended on handcrafted features and heuristic-based rules [5], [6], [7]. However, these conventional approaches often struggle with the complexity and variability inherent in medical images [8], [9]. In recent years, with the rapid

advancement of deep learning, data-driven methods have emerged as the dominant solutions for medical image segmentation tasks.

In recent years, U-Net [10] and its variants have achieved remarkable success in the field of medical image segmentation, significantly advancing research and practical applications. The original U-Net demonstrated the effectiveness of encoder-decoder convolutional architectures with skip connections in biomedical image segmentation. Building upon this foundation, numerous improvements have emerged. For instance, UNet++ [11] introduces a series of nested encoder-decoder sub-networks to redesign the skip connections, enabling more refined feature fusion across scales. UNet3+ [12] proposes full-scale skip connections

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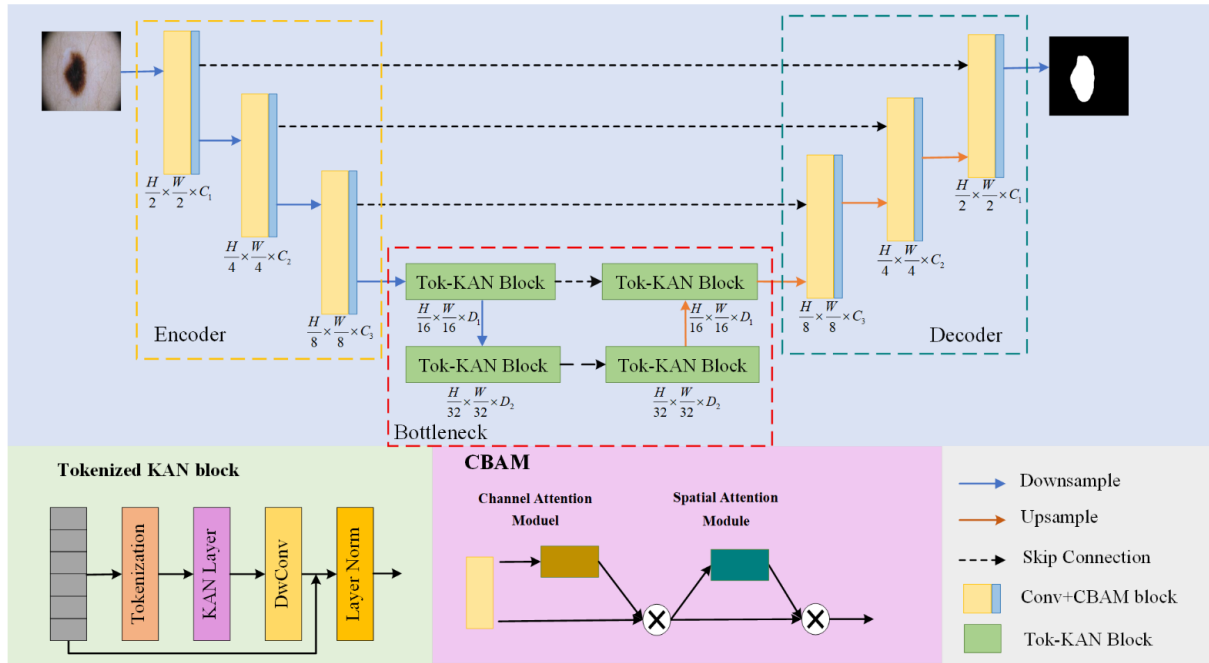


FIGURE 1. Overview of the KC-UNet architecture, which is built upon U-Net by integrating Tokenized KAN modules between the encoder and decoder, and embedding CBAM attention modules at each stage to enhance feature representation.

to directly aggregate semantic information from multi-level feature maps, while also utilizing deep supervision to enhance hierarchical representation learning. To address the complexity of 3D medical imaging, models such as 3D U-Net [13] and V-Net [14] extend the traditional U-Net by incorporating 3D convolutions, thereby enhancing volumetric segmentation performance. More recently, hybrid architectures like U-NeXt [15] have combined convolutional operations with Multi-Layer Perceptrons (MLPs), improving segmentation efficiency especially in resource-constrained scenarios.

Despite their impressive performance, conventional CNNs are inherently limited in capturing long-range dependencies due to their local receptive fields [16]. This limitation has

spurred the development of alternative architectures capable of modeling global contextual information more effectively. Among them, Transformer-based models have shown great promise in medical image segmentation by leveraging self-attention mechanisms to establish long-range dependencies and enhance feature representation robustness.

Representative models include TransUNet [17], the first framework to combine Transformers with the U-Net architecture for medical image segmentation. MedT [18] introduces a gated axial-attention mechanism designed to cope with the challenge of limited annotated data. Additionally, Swin-unet [19] employs Swin Transformers to further improve hierarchical feature modeling and efficiency. KANDU-Net [20] introduces a dual-path structure that integrates the KAN [21] and convolutional layers within the U-Net framework. TransUKAN [22] has been proposed as a hybrid framework that integrates KAN, Transformer modules, and a U-Net-like architecture, this design enables

the model to more effectively capture both fine-grained local features and long-range global dependencies, thereby improving segmentation performance.

Despite their advantages, both CNNs and Transformer-based architectures have inherent limitations [23]. CNNs are mainly effective at capturing local spatial features, making it difficult to model complex inter-channel nonlinear dependencies that are crucial for accurate diagnosis [24]. Transformers, on the other hand, often rely on large-scale labeled datasets for effective training—resources that are typically scarce in the medical domain. Moreover, Transformer-based models tend to struggle with the extraction of fine-grained local details, which are equally essential in precise lesion boundary delineation [25].

To address these limitations, this paper proposes KC-UNet, a novel architecture that integrates Tokenized Kolmogorov–Arnold Networks (Tok-KAN) and Convolutional Block Attention Module (CBAM) [26] into a unified segmentation framework. Unlike previous approaches that rely on global KAN modeling or simple MLP-based alternatives, our method introduces a token-level decomposition strategy within the encoder, enabling Tok-KAN to efficiently capture high-order, nonlinear dependencies across spatially coherent but sparse regions. This design is particularly advantageous in medical imaging, where lesion areas often exhibit irregular morphology and limited spatial extent. Furthermore, CBAM modules are embedded throughout the encoder and decoder to guide adaptive feature refinement, while dilated convolutions enhance the model’s ability to aggregate multi-scale contextual information. This synergy leads to a lightweight yet expressive segmentation model,

demonstrating superior performance and generalizability across diverse medical imaging tasks.

II. METHODS

A. OVERVIEW ARCHITECTURE

Fig. 1 illustrates the overall architecture of the proposed KC-UNet model. Built upon the classical U-Net framework, KC-UNet preserves the symmetric encoder–decoder structure while integrating two key components: Tok-KAN modules and CBAM, aiming to enhance global modeling capacity and adaptive feature selection. To further strengthen the model's focus on informative regions, CBAM is embedded within both the encoder and decoder stages. The Channel Attention Module (CAM) and Spatial Attention Module (SAM) are applied sequentially, enabling fine-grained recalibration of feature maps [27]. In the encoder path, the input image is first processed through a series of convolutional layers with embedded CBAM to extract hierarchical features. The resulting feature maps are then tokenized and fed into the Tok-KAN modules [28], which consist of multiple stacked blocks designed to capture complex nonlinear interactions and long-range dependencies. In the decoder, spatial resolution is progressively restored via up-sampling, while skip connections from the encoder facilitate the fusion of high-level semantic and low-level spatial features. As a whole, this design leads to more precise and semantically rich segmentation outputs.

B. CONVOLUTION PHASE

Each convolutional block in our network consists of three essential components: a convolution layer (Conv), batch normalization (BN), and the ReLU activation. A 3×3 kernel is used with a stride of 1 and padding of 1, allowing the feature map's spatial resolution to remain unchanged post-convolution. To further normalize the output and accelerate the convergence during training, batch normalization is applied immediately after convolution. Subsequently, a ReLU activation function introduces non-linearity into the network, enabling it to learn more complex feature representations.

In the encoder pathway, each convolutional block is additionally coupled with a 2×2 max-pooling layer. Formally, given an image $X_0 = I \in R^{H_0 \times W_0 \times C_0}$, the output of each convolution block can be elaborated as:

$$X_l = Pool (Conv(X_{l-1})) \quad (1)$$

where $X_l = I \in R^{H_l \times W_l \times C_l}$ denotes the output feature maps at the l -th layer. Given the configuration that there are L blocks in the Convolutional Phase, the final output is derived as X_L .

C. CBAM

To enhance the model's feature extraction capabilities, this study integrates the Convolutional Block Attention Module (CBAM) into both the encoder and decoder of the U-Net architecture. CBAM is a lightweight and modular attention

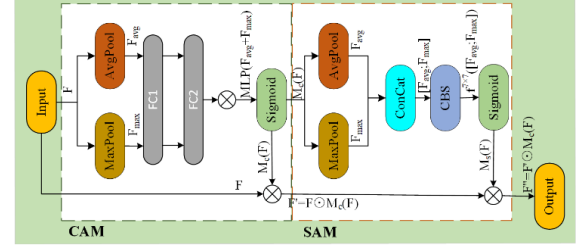


FIGURE 2. Structure of CBAM module.

mechanism that refines feature maps along both the channel and spatial.

As shown in **Fig. 2**, CBAM sequentially applies Channel Attention Module (CAM) and Spatial Attention Module (SAM). CAM utilizes global average and max pooling followed by a shared MLP to generate channel-wise attention weights, emphasizing semantically relevant channels such as edge or texture-related responses. The channel attention map $M_C \in R^{C \times 1 \times 1}$ is computed as:

$$M_C(F) = \sigma (MLP(F_{avg}) + MLP(F_{max})) \quad (2)$$

where $F_{avg} = AvgPool(F)$, $F_{max} = MaxPool_{ch}(F)$, $F \in R^{C \times H \times W}$ is the input feature map, σ represents the sigmoid activation function, while MLP refers to a shared multi-layer perceptron module.

SAM then generates spatial attention by compressing along the channel axis and highlighting spatial regions likely to contain lesions or structural features. The spatial attention map $M_S \in R^{1 \times H \times W}$ is computed as:

$$M_S(F) = \sigma (f^{7 \times 7} ([F_{avg}; F_{max}])) \quad (3)$$

where $F_{avg} = AvgPool_{ch}(F')$, $F_{max} = MaxPool_{ch}(F')$, $F' \in R^{C \times H \times W}$ is the input feature map enhanced by channel attention, $f^{7 \times 7}$ signifies a convolution operation with a 7×7 kernel.

For both encoder and decoder blocks, let F_l denote the input feature map of the l -th convolutional block. CBAM is applied as:

$$F'_l = M_S(M_C(F_l)) \odot F_l \quad (4)$$

In the encoder, F_l comes from the convolutional block output before downsampling. This allows early suppression of irrelevant background and emphasizes discriminative object regions.

In the decoder, F_l comes from the convolutional block output after skip connection fusion, guiding selective enhancement of relevant features while suppressing noisy activations.

D. TOKENIZED KAN PHASE

This study adopts a Tokenized KAN structure following the design proposed in Li et al. [29], with slight modifications to fit our architecture. Liu et al. introduced the Kolmogorov-Arnold Network (KAN), drawing inspiration from the Kolmogorov-Arnold representation theorem.

The embedded tokens Z_0 are subsequently passed through a stack of K KAN layers. The entire KAN block is represented as a nested composition of kernel-based mappings:

$$KAN(Z) = (\vartheta_{K-1}^\circ \vartheta_{K-2}^\circ \cdots \vartheta_1^\circ \vartheta_0) Z \quad (5)$$

Each layer ϑ_k maps input dimensionality n_{in} to output dimensionality n_{out} , parameterized by a set of learnable kernel activation functions:

$$\vartheta = \{\vartheta_{q,p}\}, p = 1, 2, \dots, n_{in}, q = 1, 2, \dots, n_{out} \quad (6)$$

The transformation at layer k -th can be expressed in matrix form as:

$$Z_{k+1} = \vartheta_K(Z_k) = \begin{bmatrix} \vartheta_{K,1,1}(\cdot) & \cdots & \vartheta_{K,1,n_K}(\cdot) \\ \vdots & \ddots & \vdots \\ \vartheta_{K,n_{K+1},1}(\cdot) & \cdots & \vartheta_{K,n_{K+1},n_K}(\cdot) \end{bmatrix} Z_k \quad (7)$$

Unlike traditional MLPs that rely on static linear weights, KAN leverages flexible, parameterized nonlinear activation kernels, enhancing both representation capacity and interpretability while maintaining a relatively low parameter count.

1) TOKENIZATION

In the tokenized KAN block, the output feature maps from the convolutional stage, denoted X_l are first transformed into a sequence of flattened 2D patches $\{X_l^i \in R^{P^2 \cdot C_l} | i = 1, \dots, N\}$ through a tokenization process, where each patch has a fixed spatial size $P \times P$ and $N = H_l \times W_l / P^2$ represents the total number of patches. These vectorized patches are subsequently projected $E \in R^{(P^2 \cdot C_l) \times D}$ into a latent embedding space of dimension D via a learnable linear transformation, formulated as:

$$Z_0 = [X_l^1 E; X_l^2 E; \dots; X_l^N E] \quad (8)$$

The linear projection $E \in R^{(P^2 \cdot C_l) \times D}$ is implemented by a convolution layer with a kernel size of 3×3 , which providing both feature extraction and dimension lifting capabilities.

2) EMBEDDING BY KAN LAYER

Given the obtained tokens, we pass them into a series of KAN layers (typically $K=3$). After each KAN layer, the features are sequentially processed by an efficient depthwise convolutional layer (DwConv) [30], batch normalization (BN), and a ReLU activation function. A residual connection is employed in this stage by incorporating the original token representations as additive residuals. We then apply a layer normalization (LN) [31] and pass the output features to the next block. Formally, the output of k -th Tokenized KAN block can be elaborated as:

$$Z_k = LN(Z_{k-1} + DwConv(KAN(Z_{k-1}))) \quad (9)$$

where Z_k is the output feature maps at k -th layer. Given the setup that there are K blocks in the Tokenized KAN Phrase, the final output is derived as Z_k .

E. EVALUATION INDICATORS

The Intersection over Union (IoU) metric is widely used in object and image segmentation to evaluate the alignment between predicted and ground truth regions. It is the ratio of the overlap area to the union area of the two. IoU ranges from 0 to 1, with 1 indicating perfect overlap and optimal segmentation. In medical ultrasound image segmentation, a higher IoU denotes more accurate delineation of target structures. It is computed as:

$$IOU = \frac{TP}{TP + FP + FN} \quad (10)$$

The Dice Similarity Coefficient (DSC) is a commonly used metric for evaluating the performance of image segmentation tasks, particularly in medical imaging. It measures the spatial overlap between the predicted segmentation and the ground truth annotation, with values ranging from 0 to 1. A DSC value of 1 indicates complete agreement between the two masks, reflecting ideal segmentation performance, whereas a value of 0 indicates no overlap, corresponding to a failed segmentation. The mathematical formulation of DSC is expressed as follows:

$$Dice = \frac{2TP}{2TP + FP + FN} \quad (11)$$

Accuracy (ACC) is a fundamental metric used to evaluate the overall correctness of a segmentation or classification model. It is defined as the proportion of correctly predicted pixels, including both foreground and background, to the total number of pixels. Although accuracy provides a general assessment of performance, it may be less informative in cases of class imbalance. The metric is calculated as:

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \quad (12)$$

Recall, also referred to as sensitivity or true positive rate, measures the proportion of actual foreground pixels (positive class) that are correctly identified by the model. It is particularly important in medical image segmentation, where missing pathological regions may have serious clinical consequences. A higher recall indicates fewer false negatives and better lesion detection capability. The recall is defined as:

$$Recall = \frac{TP}{TP + FN} \quad (13)$$

In the context of pixel-level segmentation, True Positives (TP) refer to the count of foreground pixels accurately identified by the model, while True Negatives (TN) represent background pixels that are correctly classified. False Positives (FP) denote background pixels erroneously labeled as foreground, and False Negatives (FN) correspond to foreground pixels that are mistakenly classified as background.

III. EXPERIMENTAL RESULTS

This study conducted a comprehensive evaluation of the proposed algorithm on four distinct and heterogeneous datasets, namely BUSI, GLAS, CVC, and ISIC2017. Each dataset possesses unique characteristics, varying data volumes, and different image resolutions. These datasets are commonly used in image segmentation, providing a reliable testing platform to assess the effectiveness and adaptability of our method.

A. DATASETS

The BUSI dataset [32] includes ultrasound images of normal, benign, and malignant breast tumors with segmentation masks. This study used a subset of 631 benign and malignant cases for experiments. The dataset offers diverse tumor presentations, supporting automated diagnosis and breast cancer research.

The CVC-ClinicDB dataset [33], commonly referred to as “CVC,” is a publicly available resource for diagnosing polyps in colonoscopy videos. The dataset comprises 612 images with a fixed resolution of 384×288 , meticulously curated from 31 distinct colonoscopy video sequences. These images capture a variety of polyp cases, making the dataset valuable for developing and accessing polyp detection algorithms.

The GLAS dataset [34] comprises 165 annotated colorectal histology images of varying resolutions (approximately 775×522 to 1512×1016). It is a widely used benchmark for gland segmentation. Following standard practice, all images were resized to 256×256 pixels in this study.

The ISIC2017 Dataset [35], the International Skin Imaging Collaboration 2017 dataset is a publicly available benchmark for skin lesion segmentation, consisting of 2,150 dermoscope images with corresponding segmentation masks.

Prior to the training of the model, a systematic preprocessing pipeline was implemented for the medical images. Specifically, median filtering was utilized to mitigate random noise, while adaptive histogram equalization (CLAHE) was employed to enhance image contrast. Pixel values were normalized to a range of $[0,1]$, and all images were uniformly resized to 256×256 pixels in order to meet the input requirements of the model. To further enhance model generalization, data augmentation strategies such as random flipping and rotation were incorporated during the training process. These procedures effectively improved both image quality and diversity, thereby providing a robust foundation for subsequent model training. The comparison between original images and preprocessed images is illustrated in Fig. 3.

B. IMPLEMENTATION DETAILS

All experiments were conducted using 5-fold cross-validation [36] to ensure robust and unbiased model evaluation. For datasets with explicit patient identifiers (e.g., BUSI and GLAS), a strict patient-level split was enforced within each fold, ensuring that all images from the same

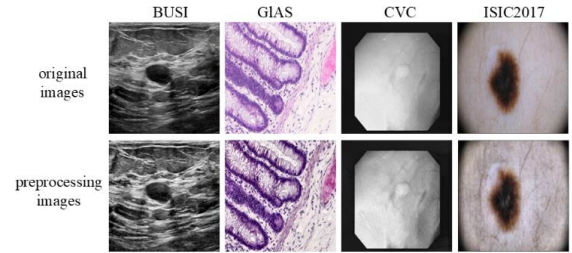


FIGURE 3. Comparison of original and preprocessed images from different medical image datasets.

patient appeared exclusively in either the training or validation set. For datasets lacking patient IDs (e.g., ISIC2017 and CVC), samples were randomly partitioned at the image level for 5-fold cross-validation, consistent with prevailing practices in the literature. Ablation studies were performed exclusively on the ISIC2017 dataset, which offers a sufficiently large and diverse set of skin lesion images. Although it does not support patient-level splitting, its wide adoption in segmentation benchmarks makes it a representative dataset for evaluating the relative contribution of each module under consistent conditions. Conducting ablation on a single dataset also helps reduce variability introduced by inter-dataset differences, allowing for clearer insight into the effectiveness of individual components. Model training was conducted using the SGD optimizer with an initial learning rate of 0.01, a minimum learning rate of $1e-5$, and weight decay to mitigate overfitting. A total of 300 training epochs were executed, with the first 10 epochs designated as a warm-up phase using linear learning rate scheduling, followed by cosine annealing. The batch size was set to 8. The loss function comprised a weighted combination of cross-entropy loss and Dice loss. Performance evaluation was conducted using Dice, IoU, Acc, and Recall, while model parameters and inference counts were recorded for comprehensive assessment. Final results were reported as the average across the five validation folds.

C. ABLATION STUDIES

1) THE NUMBER OF KAN LAYER

To evaluate the impact of KAN layer depth on segmentation performance, we varied the number of KAN layers in the Tokenized KAN phase from one to five, while keeping all other architectural components and training configurations fixed. Table 1 summarizes the quantitative results averaged over five cross-validation folds. The model with three KAN layers achieved the highest IoU (83.25%) and Dice score (90.69%), whereas both shallower (1 or 2 layers) and deeper (4 or 5 layers) configurations led to inferior performance. To assess whether the performance differences were statistically significant, paired t-tests were conducted on the cross-validation results. The 3-layer KAN configuration significantly outperformed all others in terms of both IoU and Dice ($p < 0.05$), indicating that this setting offers an optimal balance between representation capacity and overfitting risk.

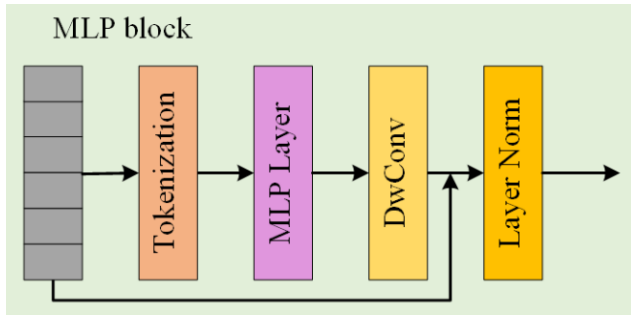
TABLE 1. Ablation studies on number of used KAN layers.

#KAN	IoU (%)	Dice (%)	Acc (%)	Recall (%)
1 Layer	82.95	89.35	89.76	91.31
2 Layer	82.07	89.30	88.94	90.40
3 Layer	83.25*	90.69*	90.14	91.83
4 Layer	82.13	89.52	89.35	91.30
5 Layer	81.13	88.92	88.81	90.29

* $p < 0.05$ vs. all other configurations (paired t-test across folds)

TABLE 2. Ablation results of replacing Tokenized KAN blocks with MLP blocks.

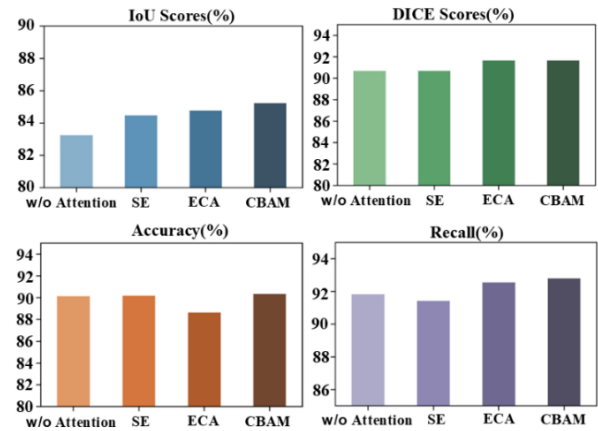
KAN vs. MLP block	IoU (%)	Dice (%)	Acc (%)	Recall (%)
MLP $\times$ 4	82.13	89.13	88.89	90.25
Random KAN ($\times$ 1)	82.28	89.17	89.36	90.43
KAN $\times$ 4	83.25	90.69	90.14	91.83

**FIGURE 4.** Structure of the MLP block, replacing the KAN Layer in the Tokenized KAN block with an MLP layer for ablation comparison.

2) REPLACING TOKENIZED KAN BLOCKS WITH MLP BLOCKS

To further evaluate the effectiveness of the proposed Tokenized KAN block, we conducted ablation experiments by replacing some or all of the Tokenized KAN blocks in the network with structurally equivalent MLP blocks. These MLP blocks retained the same architectural components—tokenization, depthwise convolution, and layer normalization—with the only modification being the substitution of the KAN layer with a standard MLP layer, implemented as a two-layer perceptron with GELU activation and a residual connection.

Fig. 4 depicts the detailed structure of the MLP block employed in our experiments. To comprehensively assess the role of Tokenized KAN blocks, we evaluated three configurations: replacing all four KAN blocks with MLP blocks (MLP $\times$ 4), retaining a single randomly selected KAN block while replacing the other three with MLP blocks (Random KAN $\times$ 1), and preserving all four blocks as Tokenized KAN (KAN $\times$ 4). Table 2, show that the full KAN configuration (KAN $\times$ 4) outperforms the others, achieving the highest IoU (83.25%), Dice score (90.69%), accuracy (90.14%), and recall (91.83%). The Random KAN ($\times$ 1) configuration yields intermediate performance (IoU 82.28%, Dice 89.17%), slightly surpassing the fully replaced MLP $\times$ 4 setup (IoU 82.13%, Dice 89.13%), indicating that even

**FIGURE 5.** Performance comparison of different attention mechanisms on ISIC2017 dataset.**TABLE 3.** KC-UNet performance with different attention mechanisms on ISIC2017.

Attention Module	IoU (%)	Dice (%)	Acc (%)	Recall (%)
Base +w/o Attention	83.25	90.69	90.14	91.83
Base +SE	84.48	91.46	90.18	91.43
Base +ECA	84.77	91.66	88.63	92.55
Base +CBAM	85.23	92.11	90.34	92.79

partial retention of KAN blocks benefits segmentation performance.

3) COMPARISON OF ATTENTION MECHANISMS

To verify the effectiveness of the attention mechanisms integrated into the proposed KC-UNet architecture, a series of ablation experiments were conducted. The base model consists of U-Net integrated with KAN. Building upon this, three widely used attention modules, CBAM, SE [37], and ECA [38], were evaluated to assess their impact on segmentation performance. Each module was inserted at identical locations within the network, specifically after every encoder and decoder layer, to ensure a fair comparison.

Table 3 and Fig. 5 present a quantitative and visual comparison of each model in terms of Dice coefficient, IoU, Accuracy, and Recall. Compared with the baseline model without attention mechanisms, introducing attention modules led to varying degrees of performance improvement. Among them, the CBAM module achieved an increase of 1.42% in Dice coefficient, 1.98% in IoU, 0.20% in Accuracy, and 0.96% in Recall. In contrast, the SE and ECA modules also brought performance gains, but the overall improvements were slightly lower than those of CBAM, with the Dice coefficient increasing by 0.77% and 0.97%, respectively.

Overall, CBAM by jointly modeling channel and spatial attention, significantly enhanced the model's ability to perceive critical regions and express features. Therefore, CBAM was ultimately selected as the attention mechanism component for the UKAN model in this study.

D. PERFORMANCE COMPARISON ON IMAGE SEGMENTATION

Tables 4 and Tables 5 report the quantitative performance of KC-UNet in comparison with several representative segmentation models across four benchmark medical image datasets. The evaluation is based on two widely adopted metrics: IoU and Dice.

As summarized in Table 4, KC-UNet achieves the highest IoU scores across all datasets. Specifically, it records an IoU of 66.60% on BUSI, outperforming the convolutional baseline U-Net (59.21%) and even the transformer-based Swin-UNet (65.32%). On GLAS, KC-UNet attains 87.96%, improving upon U-KAN (87.28%), TransUNet (80.74%), and U-Net (85.13%). Similarly, for CVC and ISIC2017, KC-UNet achieves 82.59% and 85.24%, respectively, consistently leading all competing methods.

The Dice coefficient results in Table 5 further confirm the advantage of KC-UNet. It achieves 80.46% on BUSI, surpassing Swin-UNet (78.21%) and TransUNet (76.59%). On GLAS, it reaches 92.85%, slightly outperforming U-KAN (92.15%). For CVC (92.25%) and ISIC2017 (92.11%), KC-UNet achieves the best DSC values across the board.

TABLE 4. IoU results (%) on four medical image datasets (mean $\pm$ standard deviation).

Methods	BUSI	GLAS	CVC	ISIC2017
U-Net	59.21 $\pm$ 0.42	85.13 $\pm$ 0.33	80.92 $\pm$ 0.18	80.42 $\pm$ 0.04
Att-UNet	58.94 $\pm$ 0.29	85.06 $\pm$ 0.44	81.04 $\pm$ 0.23	80.13 $\pm$ 0.10
TransUNet	61.62 $\pm$ 0.41	80.74 $\pm$ 0.20	82.17 $\pm$ 0.19	84.48 $\pm$ 0.05
Swin-unet	65.32 $\pm$ 0.32	83.34 $\pm$ 0.17	82.02 $\pm$ 0.31	85.01 $\pm$ 0.10
U-KAN	64.07 $\pm$ 0.24	87.28 $\pm$ 0.10	81.47 $\pm$ 0.21	84.03 $\pm$ 0.08
KC-UNet	66.60$\pm$0.16	87.96$\pm$0.08	82.59$\pm$0.18	85.24$\pm$0.03

TABLE 5. Dice Coefficient results (%) on four medical image datasets (mean $\pm$ standard deviation).

Methods	BUSI	GLAS	CVC	ISIC2017
U-Net	73.02 $\pm$ 0.12	89.88 $\pm$ 0.08	88.71 $\pm$ 0.14	89.12 $\pm$ 0.06
Att-UNet	72.14 $\pm$ 0.19	90.51 $\pm$ 0.41	88.19 $\pm$ 0.13	88.58 $\pm$ 0.11
TransUNet	76.59 $\pm$ 0.41	87.11 $\pm$ 0.33	91.70 $\pm$ 1.25	89.25 $\pm$ 0.09
Swin-unet	78.21 $\pm$ 0.63	88.23 $\pm$ 0.20	91.03 $\pm$ 0.25	91.50 $\pm$ 0.29
U-KAN	77.89 $\pm$ 0.28	92.15 $\pm$ 0.15	89.97 $\pm$ 0.18	91.07 $\pm$ 0.08
KC-UNet	80.46$\pm$0.11	92.85$\pm$0.08	92.25$\pm$0.15	92.11$\pm$0.02

To offer a more intuitive comparison, we present two scatter plots in Fig. 6, illustrating the performance distribution of all models across the datasets in terms of IoU and DICE, respectively. Each point represents the performance of a method on a specific dataset. As shown, KC-UNet consistently clusters in the upper-right corner in both plots, indicating both higher accuracy and greater stability. This visual analysis clearly confirms that KC-UNet exhibits not only top-level performance but also strong generalization across different medical imaging modalities.

The superior performance observed across all datasets is primarily attributed to the architectural design of KC-UNet. In particular, the integration of KAN enhances the model's capacity to capture long-range dependencies more effectively than traditional convolutional or

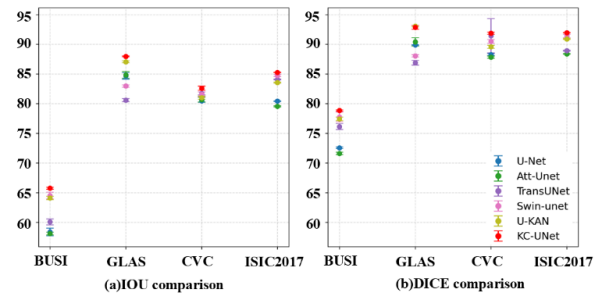


FIGURE 6. Performance comparison of different attention mechanisms on ISIC2017 dataset.

TABLE 6. Model efficiency and inference time.

Methods	Efficiency		
	GFLOPS(G)	Params (M)	Inference time(ms/img)
U-Net	54.77	31.05	17.99
Att-UNet	55.85	31.40	18.43
TransUNet	32.24	105.30	28.70
Swin-unet	8.69	41.38	26.69
U-KAN	7.09	10.19	8.94
KC-UNet	7.10	10.22	12.63

transformer-based components. Furthermore, the incorporation of the CBAM contributes to improved feature discrimination by adaptively refining both spatial and channel-wise representations. These enhancements collectively enable KC-UNet to achieve more accurate and robust segmentation performance across diverse medical imaging tasks.

Table 6 provides a comparative analysis of computational efficiency, model complexity, and inference speed across the evaluated segmentation networks, reported as GFLOPS, parameter count (in millions), and per-image inference time (ms/img). The proposed KC-UNet achieves a balanced trade-off between performance and computational cost. While Transformer-based architectures such as TransUNet (32.24 G, 105.30 M, 28.70 ms/img) and Swin-Unet (8.69 G, 41.38 M, 26.69 ms/img) deliver strong segmentation accuracy, they incur higher memory demands and longer inference times.

In contrast, KC-UNet requires only 7.10G and 10.22 M parameters, with an inference time of 12.63 ms/img, representing a substantial reduction in computational load compared to TransUNet and Swin-Unet. This efficiency is achieved while maintaining competitive performance, being only marginally heavier than U-KAN (7.09 G, 10.19 M, 8.94 ms/img) from which it is derived. Moreover, compared with conventional CNN-based architectures such as U-Net (54.77 G, 31.05 M, 17.99 ms/img) and Att-UNet (55.85 G, 31.40 M, 18.43 ms/img), KC-UNet demonstrates markedly lower computational requirements and faster inference, while retaining strong representational capability. These results underscore the suitability of KC-UNet for deployment in real-time and resource-constrained medical image segmentation tasks.

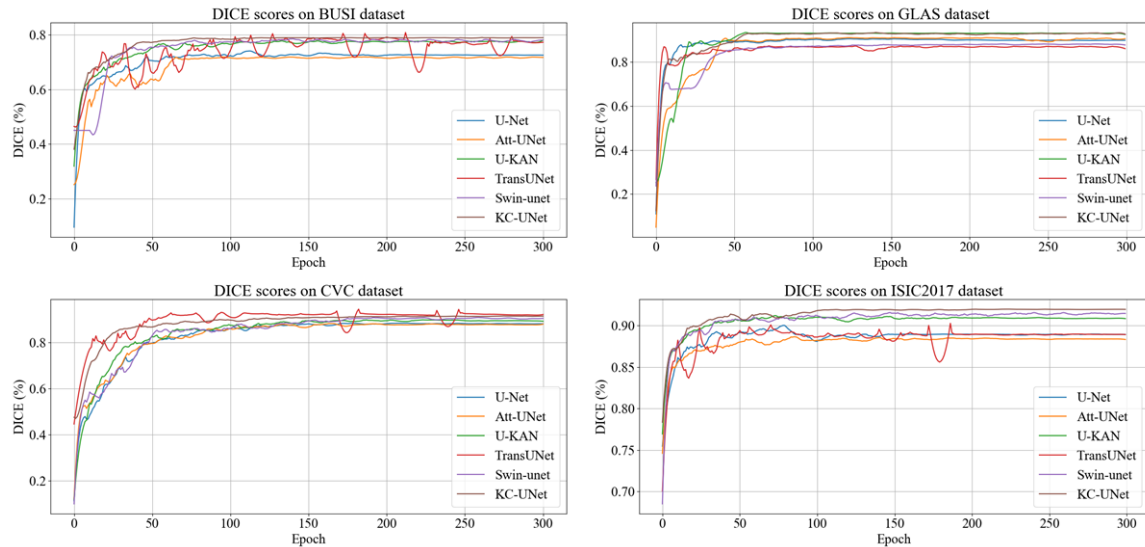


FIGURE 7. Dice comparison of different models across four datasets.

As illustrated in **Fig. 7**, KC-UNet consistently demonstrates superior segmentation performance across four representative medical image datasets, as evidenced by DICE scores throughout the training epochs. On the GLAS and CVC datasets—characterized by relatively clear glandular boundaries and moderate anatomical complexity—KC-UNet not only converges rapidly but also achieves higher final accuracy compared to both CNN-based models (U-Net, Att-UNet, U-KAN) and Transformer-based approaches (Swin-unet, TransUNet).

In the case of the BUSI dataset, which presents greater challenges due to small-scale structures and blurred lesion contours, KC-UNet exhibits remarkable robustness and maintains stable training dynamics over time, ultimately outperforming all baseline models. Similarly, on the ISIC2017 dataset—which is characterized by large-scale lesions and significant inter-class variability KC-UNet retains its performance advantage, demonstrating faster convergence and achieving higher DICE scores than competing models.

As shown in **Fig. 8**, we conducted a comprehensive qualitative comparison across all datasets. The experimental results indicate that traditional CNN-based methods such as U-Net and Att-UNet are prone to over-segmentation or under-segmentation in complex scenarios, highlighting their limitations in global feature modeling and detail preservation. Transformer-based models like TransUNet and Swin-UNet show improvements in capturing contextual information; however, they often produce blurry or fragmented boundaries, especially in datasets with small lesions or fine-grained structures.

In contrast, our proposed KC-UNet demonstrates superior visual performance. On datasets such as BUSI and GLAS, it provides more accurate delineation of lesion contours and better suppression of background noise. Notably, for

challenging cases in ISIC2017, KC-UNet excels in maintaining structural integrity and preserving subtle morphological details. These visual results strongly validate the effectiveness of our architecture design, particularly the integration of CBAM and KAN modules in enhancing both local precision and global coherence.

E. RESULTS

The experimental evaluation was conducted on four heterogeneous and representative medical image segmentation datasets: BUSI, GLAS, CVC, and ISIC2017. The proposed KC-UNet consistently demonstrates superior performance in both segmentation accuracy and generalization capability. Quantitative comparisons using IoU and Dice coefficients show that KC-UNet outperforms classical CNN-based models such as U-Net and Att-UNet, improved CNN architectures like U-KAN, and state-of-the-art Transformer-based methods including TransUNet and Swin-UNet.

Specifically, KC-UNet achieves the highest IoU scores across all four datasets: 66.60% on BUSI, 87.96% on GLAS, 82.59% on CVC, and 85.24% on ISIC2017. In terms of Dice coefficient, KC-UNet also consistently ranks at the top, achieving 80.46%, 92.85%, 92.25%, and 92.11% on the same datasets, respectively. Notably, KC-UNet surpasses Swin-UNet (IoU: 65.32%, 83.34%, 82.02%, 85.01%; Dice: 78.21%, 88.23%, 91.03%, 91.50%) and TransUNet (IoU: 61.62%, 80.74%, 82.17%, 84.48%; Dice: 76.59%, 87.11%, 91.70%, 89.25%) on all benchmarks, highlighting its robust generalization and precise segmentation capability.

In terms of computational efficiency (Table 6), KC-UNet requires 7.10 G, 10.22 M, and 12.6 ms/img, representing a substantial reduction compared with TransUNet (32.24 G, 105.30 M, 28.70 ms/img) and Swin-UNet (8.69 G, 41.38 M, 26.69 ms/img). Its complexity is only marginally higher

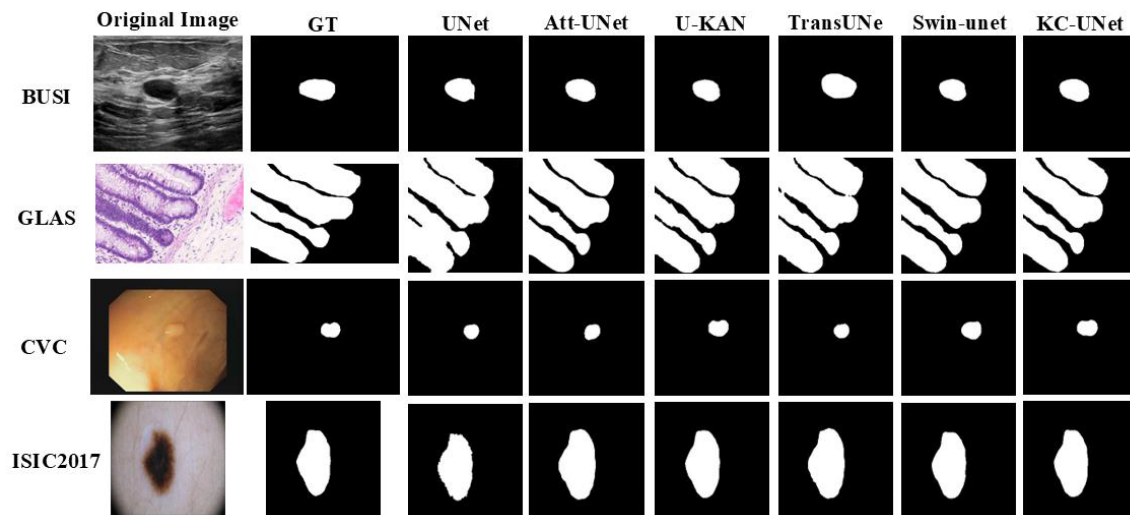


FIGURE 8. The segmentation performance in all datasets.

than that of U-KAN (7.09 G, 10.19 M, 8.94 ms/img) while delivering better segmentation accuracy. Compared with conventional CNN-based architectures such as U-Net (54.77 G, 31.05 M, 17.99 ms/img) and Att-UNet (55.85 G, 31.40 M, 18.43 ms/img), KC-UNet also demonstrates markedly lower computational requirements and faster inference.

These results collectively demonstrate that KC-UNet not only achieves state-of-the-art segmentation performance across diverse datasets, but also ensures high efficiency, making it highly suitable for deployment in real-time and resource-constrained clinical settings.

IV. CONCLUSION AND FUTURE WORKS

The KC-UNet framework proposed in this study demonstrates significant performance improvements in medical image segmentation tasks. Ablation experiments show that the application of CBAM significantly enhances the model's ability to capture complex details in medical images, particularly improving segmentation accuracy in lesion regions. Compared to other models such as Att-UNet and U-KAN, KC-UNet achieves higher IoU and DICE scores across multiple medical image datasets, with improvements ranging from 2% to 8%. Specifically, CBAM, by combining channel and spatial attention mechanisms, enables KC-UNet to more accurately focus on key regions of the image while suppressing irrelevant areas, thereby significantly improving the model's balance in extracting both local and global features.

Although KC-UNet has achieved remarkable results in segmentation accuracy, its high computational complexity remains a challenge. Especially when processing large-scale medical image data, the computational overhead of the KAN and CBAM modules cannot be ignored. Future work can focus on optimizing the computational efficiency of these modules and designing more lightweight attention mechanisms and feature extraction strategies to improve the real-time performance and usability of the model in clinical

applications. Additionally, with the continuous advancement of deep learning technologies, reducing model computation costs and parameter numbers while maintaining accuracy will become an important direction for future research.

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